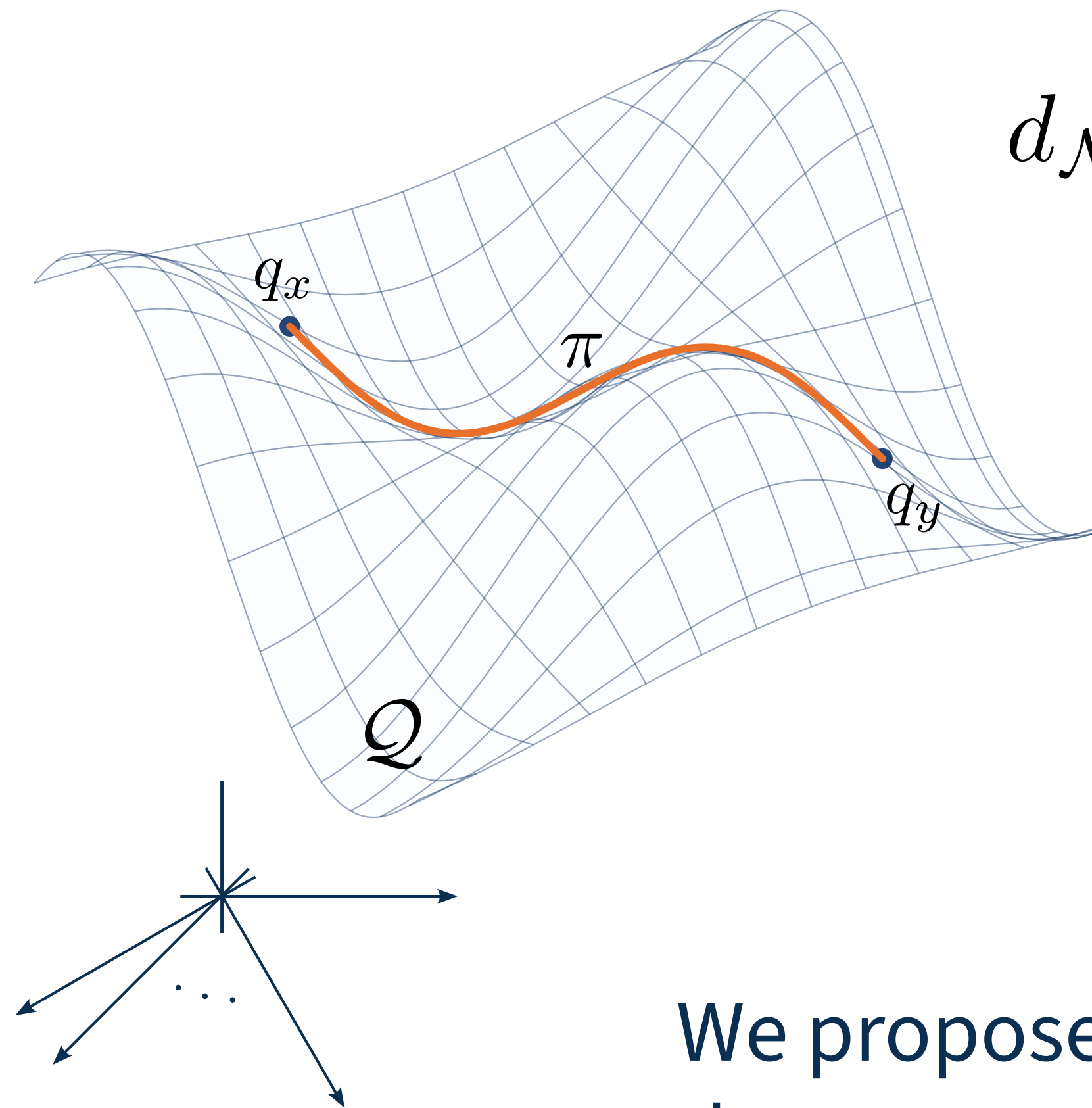




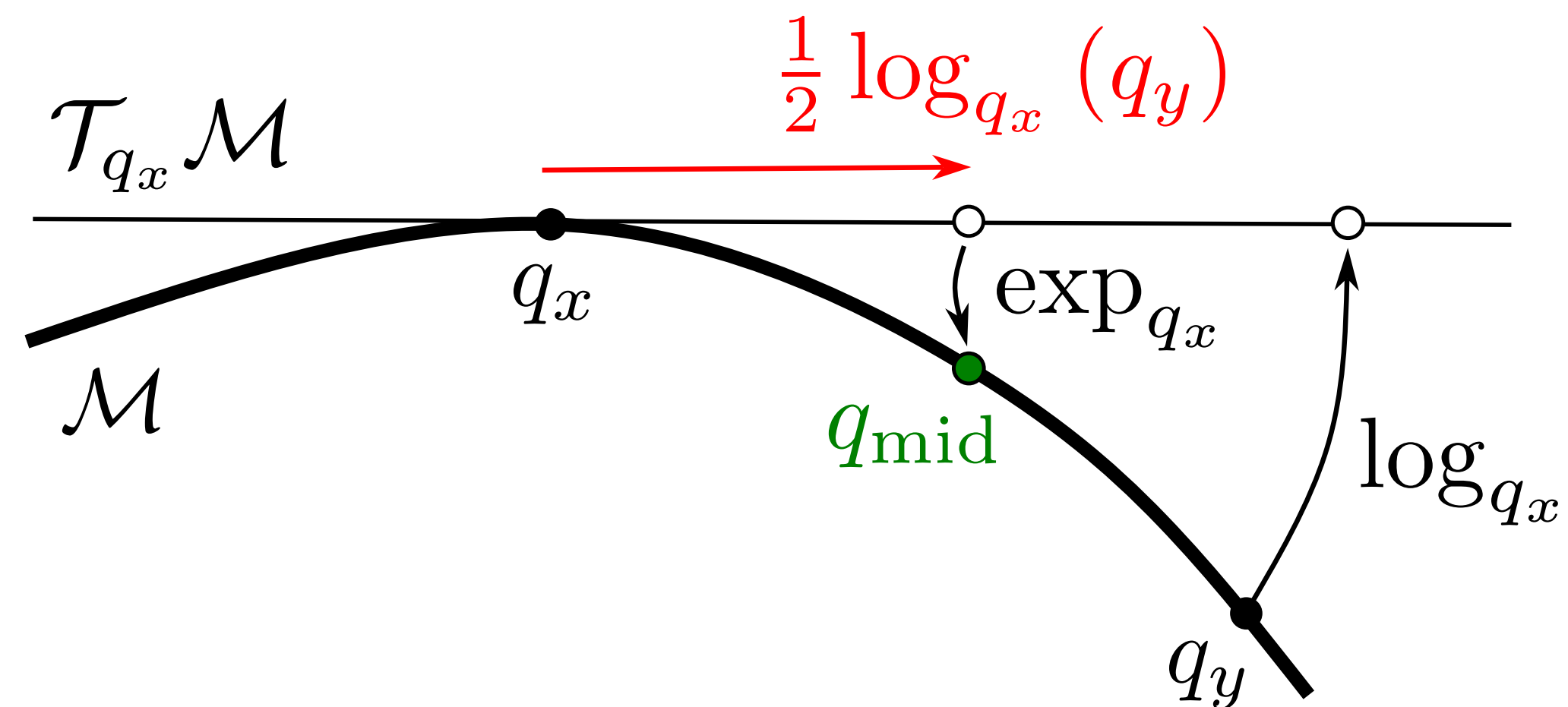
$$d_{\mathcal{M}}(q_x, q_y) = \min_{\pi} \int_0^1 \sqrt{\dot{\pi}(t)^T \underbrace{G(\pi(t))}_{\text{Riemannian metric}} \dot{\pi}(t)} dt$$

computing this *exactly* requires solving a **two-point boundary-value problem**

We propose a *lightweight* approximation that recovers the **true geodesic distance** when paired with asymptotically optimal planners like RRT*

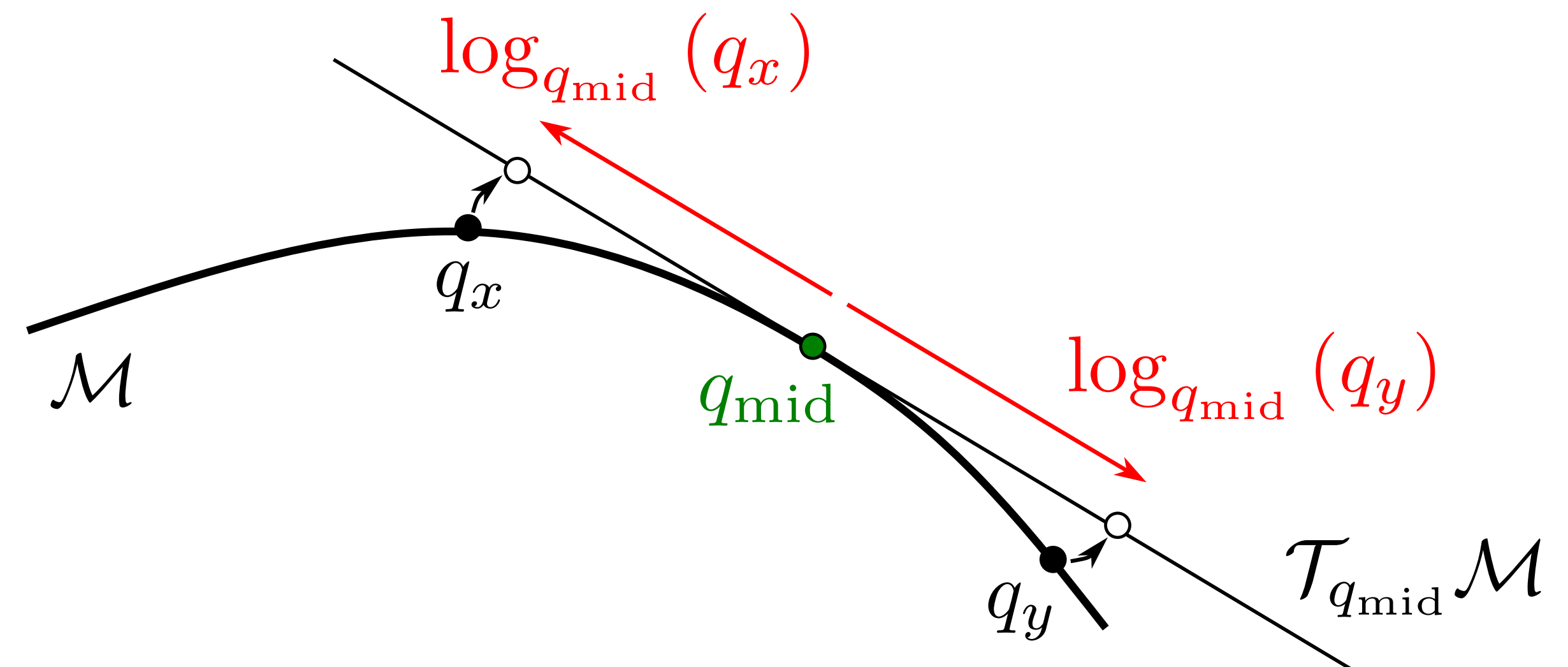
Midpoint configuration

$$q_{\text{mid}} = \exp_{q_x} \left(\frac{1}{2} \log_{q_x} (q_y) \right)$$



Riemannian distance — exact

$$d_{\mathcal{M}}(q_x, q_y) = \left\| \log_{q_{\text{mid}}}(q_y) - \log_{q_{\text{mid}}}(q_x) \right\|_{q_{\text{mid}}}$$



Replace the exp / log maps with **retractions** $\mathcal{R}_q(v), \mathcal{R}_q^{-1}(v)$

first-order • **cheap** to compute